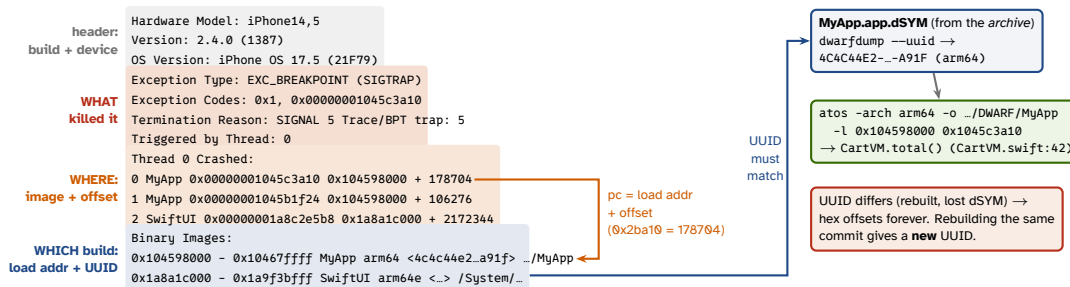


In one line: A crash report says **what** killed the process (*Exception Type + Termination Reason*), **where** (the crashed thread's backtrace of *image + offset*), and **which build** (the *UUID* of every binary image). Symbolication turns offsets into `function + file:line` using the **dSYM whose UUID matches** — no matching dSYM, no names, ever.



Exception type → cause → first move

Type (signal)	Meaning	First move
EXC_BAD_ACCESS (SIGSEGV / SIGBUS)	bad pointer: freed object, unowned, unsafe pointer; KERN_INVALID_ADDRESS	Zombies, ASan
EXC_BREAKPOINT (SIGTRAP)	Swift trap: ! on nil, index out of range, fatalError, precondition, as!, overflow	top app frame; the Fatal error msg
EXC_CRASH (SIGABRT)	abort(): uncaught NSError, C assert	Last Exception Backtrace
EXC_CRASH (SIGKILL) 0x8badf00d 0xdead10cc	watchdog: main thread blocked at launch / resume / suspend held a file or SQLite lock while being suspended	main stack; I/O off main beginBackgroundTask, release first
no crash log	jetsam (memory limit)	iOS 27 MetricKit .memoryException

How it works

- Reports are JSON .ips (iOS 15+). Always read in order: exception type → termination reason → crashed thread (main?) → first frame in *your* image.
- dSYM** = DWARF debug info split out of the stripped binary (DEBUG_INFORMATION_FORMAT = dwarf-with-dsym); the archive keeps it. Find one by UUID: `mdfind "com.apple.xcode-dsym-uuids == <UUID>".`
- Bitcode** is gone (deprecated in Xcode 14): the App Store no longer recompiles, so the *archive's* dSYM is the shipped one. Old bitcode builds needed *Download dSYMs* from App Store Connect.
- Where crashes arrive: Xcode **Organizer** (opt-in users, delayed, auto-symbolicated if the dSYM is known) · Crashlytics/Sentry (upload dSYMs from a build phase or CI) · **MetricKit**.
- MetricKit** (iOS 27): one `MetricManager` instance; `diagnosticReports` is an `AsyncSequence` of `DiagnosticReport` (Codable); result: `.crash · .hang · .cpuException · .diskWriteException · .appLaunch · .memoryException` (memory-limit kills). Before: `MXMetricManager + subscriber didReceive(_)` — deprecated in 27.2. Call `stacks` come *unsymbolicated*. Diagnostics arrive immediately (iOS 15+), metrics at most daily.

Reproduce and triage

- Memory crashes: scheme → Diagnostics → **Zombie Objects** (message to a freed object), **Address Sanitizer** (use-after-free, overflow), **Thread Sanitizer** (races; not with ASan), Main Thread Checker.
- Hangs/watchdog: Time Profiler, Hangs instrument. OOM: Allocations, Memory Graph, downsample images, drop caches on memory warning.

Example

```
let metrics = MetricManager() // iOS 27: hold ONE instance
Task {
  for await report in metrics.diagnosticReports { // never throws
    switch report.result {
    case .crash, .hang, .memoryException:
      send(report) // Codable; stacks
      unsymbolicated
      default: break
    }
  }
}
```

```
dwarfdump --uuid MyApp.app/MyApp MyApp.app.dSYM # equal?
```

Traps

- Swift traps are **not** SIGABRT: on arm64 they are EXC_BREAKPOINT (SIGTRAP) (EXC_BAD_INSTRUCTION/SIGILL on Intel simulators).
- "No crash log" ≠ no crash — jetsam kills leave only a JetsamEvent log.
- 0x8badf00d is not a bug at the top frame: it is where the main thread was when time ran out.
- Optimized builds inline code: a frame may show the caller's line or skip callees.
- Never trust "I rebuilt the same commit" — different UUID, dSYM useless.

Remember

"What · Where · Which build." Type tells the *category*, the backtrace the *place*, the UUID the *dSYM*. 8badf00d = too slow, dead10cc = lock on suspend.